CST8233

NUMERICAL COMPUTING

Learning Objectives

Statistics general information

Descriptive Statistics

- Measures Of Tendency
- Measure Of Variance

Normal Distribution

Standard Normal Distribution

Population and Sample

A **population** is the entire group that you want to make some inference or conclusion about. Population in statistics can refer to anything. It can mean people, plants, countries, weather reports, any data.

A **sample** is a specific sub-group that the data is collected from. The sample is always smaller than the population.

Population	Sample
Rock Music	Top rock songs from ‘00
weather data in Canada	rainfall patterns in Ottawa
people in Ottawa	50-60 yo in Ottawa

Statistics Basic Terminology

Data consists of information coming from observations, counts, measurements, or responses.

Quantitative data are measures of values or counts and are expressed as numbers.

Qualitative data are measures of 'types' and may be represented by a name, symbol, or a number code.

Statistics is the science of collecting, organizing, analyzing, and interpreting data in order to make decisions.

Branches of Statistics

The study of statistics has two major branches: descriptive statistics and inferential statistics.

Descriptive

Involves the organization, summarization, and display of data.

Inferential

Involves using a sample to draw conclusions about a population.

Descriptive Statistics

Broken into 2 main categories

Numerical

- Mean
- Variance
- Standard deviation
- Median
- Mode
- Skew
- etc.

Graphical

- Histogram
- Boxplot
- Scatterplot
- etc.

Descriptive Statistic: How to Summarize Data

Central Tendency (or Groups' "Middle Values")

Mean

Median

Mode

Variation (or Summary of Differences Within Groups)

Range

Variance

Standard Deviation

Mean

Important measure of “central tendency”

The arithmetic average (add all the scores together, then divide by the number of scores)

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n}$$

1+1+1+1+10=14/5=2.8

10+10+10+10+0=8

Disadvantage

Means can be badly affected by outliers (data points with extreme values unlike the rest)

Outliers can make the mean a bad measure of central tendency or common experience

Median

1+1+1+1+10=14/5=2.8, median =1

10+10+10+10+0=8, median =10

The middle value when a variable's values are ranked in order; the point that divides a distribution into two equal halves.

When data are listed in order, the median is the point at which 50% of the cases are above and 50% below it.

Advantage

The median is unaffected by outliers, making it a better measure of central tendency, better describing the "typical person" than the mean when data are skewed.

Class A--IQs of 13 Students

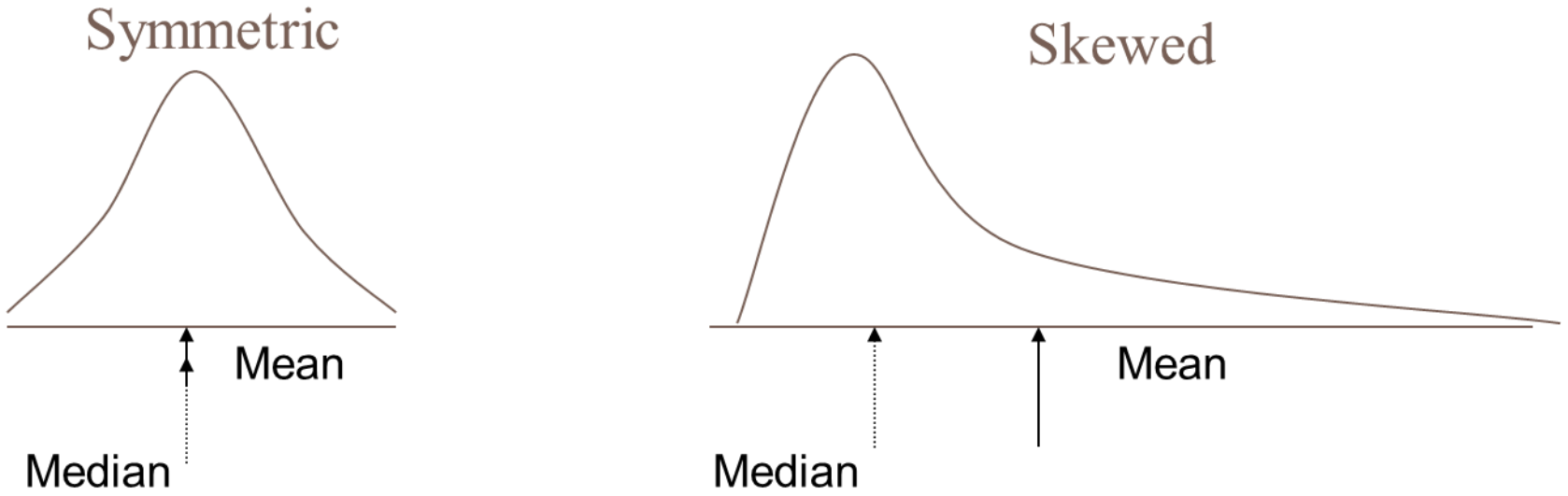
89
93
97
98
102
106
109
110
119
115
12
131
140

Median = 109

Median vs Mean

If the recorded values for a variable form a symmetric distribution, the median and mean are identical.

In skewed data, which is more common, the mean lies further toward the skew than the median.



Mode

Mode: the most common value

1, 1, 2, 4, 6, 6, 6, 7, 7, 7, 7, 8, 8, 9, 9, 10, 12, 15

It is possible to have more than one mode.

It may not be at the center of a distribution.

Variation - Range

The spread, or the distance, between the lowest and highest values of a variable. $R = y_{max} - y_{min}$

To get the range for a variable, you subtract its lowest value from its highest value.

Class A-IQs	
102	115
128	109
131	89
98	106
140	119
93	97
110	range=140-89=51

Class B-IQs	
127	162
131	103
96	111
80	109
93	87
120	108
109	range=162-80=82

Measure of Variability

Let's look at final grades for 2 different classes.

If we compute mean, we find that both classes have the same mean value, 75.

This makes comparison of data difficult.

We want to measure, or describe, the spread of the data about a central point.

Class A	Class B
75	75
80	100
70	50
77	85
73	65
75	98
90	52
60	

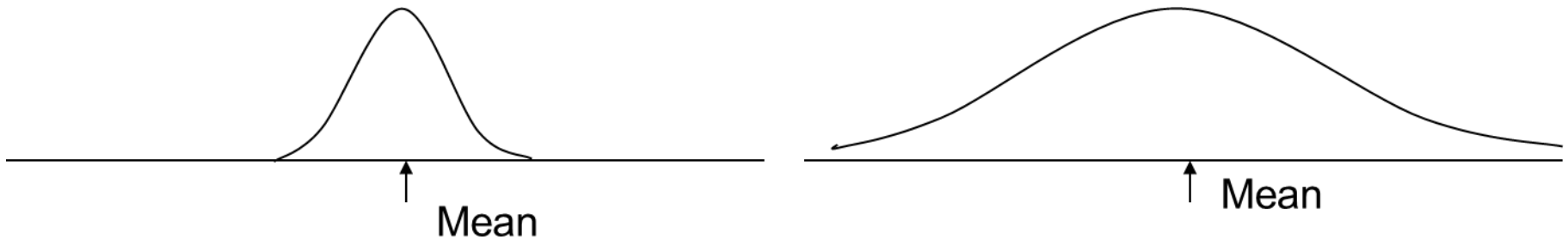
mean	mean
75	75

Variance

A measure of the spread of the data values. Variance helps to determine the data spread size.

The larger the variance, the further the individual cases are from the mean.

The smaller the variance, the closer the individual scores are to the mean.



Variance

Most important measure of “dispersion” of data.

Start with the sum of squares called Deviation - S_t

$$S_t = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Define the Sample Variance, σ as

$$\sigma^2 = \frac{S_t}{n-1} = \frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1}$$

We focus on sample statistics, so we use denominator of n-1. If you work with population statistics, you use denominator of just n. This is a very important detail to note!

Calculate the sample variance

First start with the sum of squares $S_t = \sum_{i=1}^{n}(y_i - \bar{y})^2$

$$= (75-75)^2 + (80-75)^2 + (70-75)^2 + (77-75)^2$$
$$+ (73-75)^2 + (75-75)^2 + (90-75)^2 + (60-75)^2$$
$$= 0 + 25 + 25 + 4 + 4 + 0 + 225 + 225$$
$$= 508$$

Now divide by the sample size of class A, n-1

$$= \frac{508}{7} = 72.57$$

For class B, we find that the variance is 418

Class A	Class B
75	75
80	100
70	50
77	85
73	65
75	98
90	52
60	

variance	variance
72.57	418

Variance and Standard Deviation

Based on variance we can calculate a sample standard deviation.

The square root of the variance reveals the **average deviation of the observations from the mean**.

Sample Variance = $\sigma^2 = \frac{S_t}{n-1} = \frac{\sum_{i=1}^{n}(y_i-\bar{y})^2}{n-1}$

Sample Standard Deviation =

$$\sigma = \sqrt{\left(\frac{S_t}{n-1}\right)} = \sqrt{\left(\frac{\sum_{i=1}^{n}(y_i-\bar{y})^2}{n-1}\right)}$$

Calculate the Sample Standard Deviation

Class A, $\sigma = \sqrt{72.57} = 8.51$

Class B, $\sigma = \sqrt{418} = 20.44$

With the sample mean, $\bar{y} = 75$, we can now accurately describe each data set.

Class A, $\bar{y} = 75$, σ = 8.51

Class B, $\bar{y} = 75$, σ = 20.44

Remember that the sample mean, $\bar{y}$, (sometimes $\bar{x}$) will be different from a population mean, $\bar{\mu}$

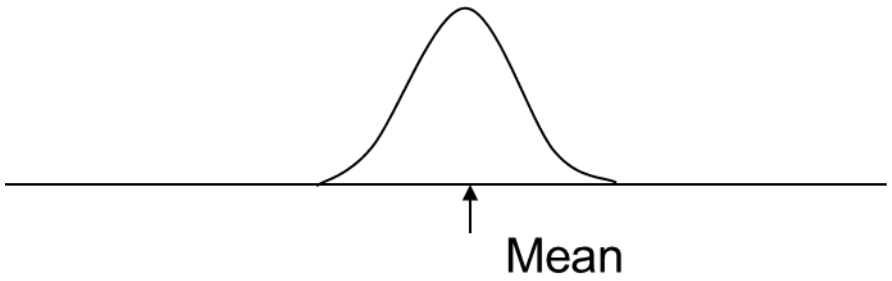
Class A	Class B
75	75
80	100
70	50
77	85
73	65
75	98
90	52
60	

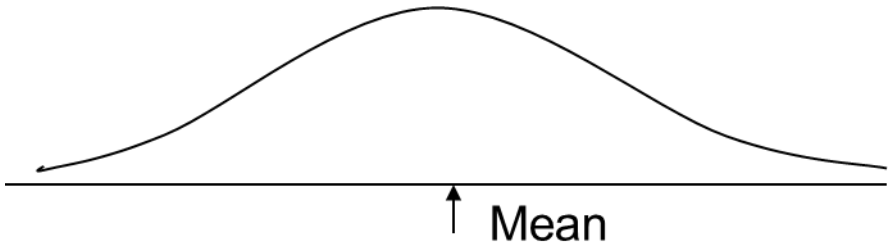
σ	σ
8.51	20.44

Variance

Class A, $\bar{y} = 75$, σ = 8.51

Class B, $\bar{y} = 75$, σ = 20.44





Your turn

Use the data in the following table to calculate:

Mean

Median

Mode

Range of data

Variance

Standard deviation

x
7
6
5
6
9

Your Turn: Solution

Mean = 6.6

Median = 6

Mode = 6

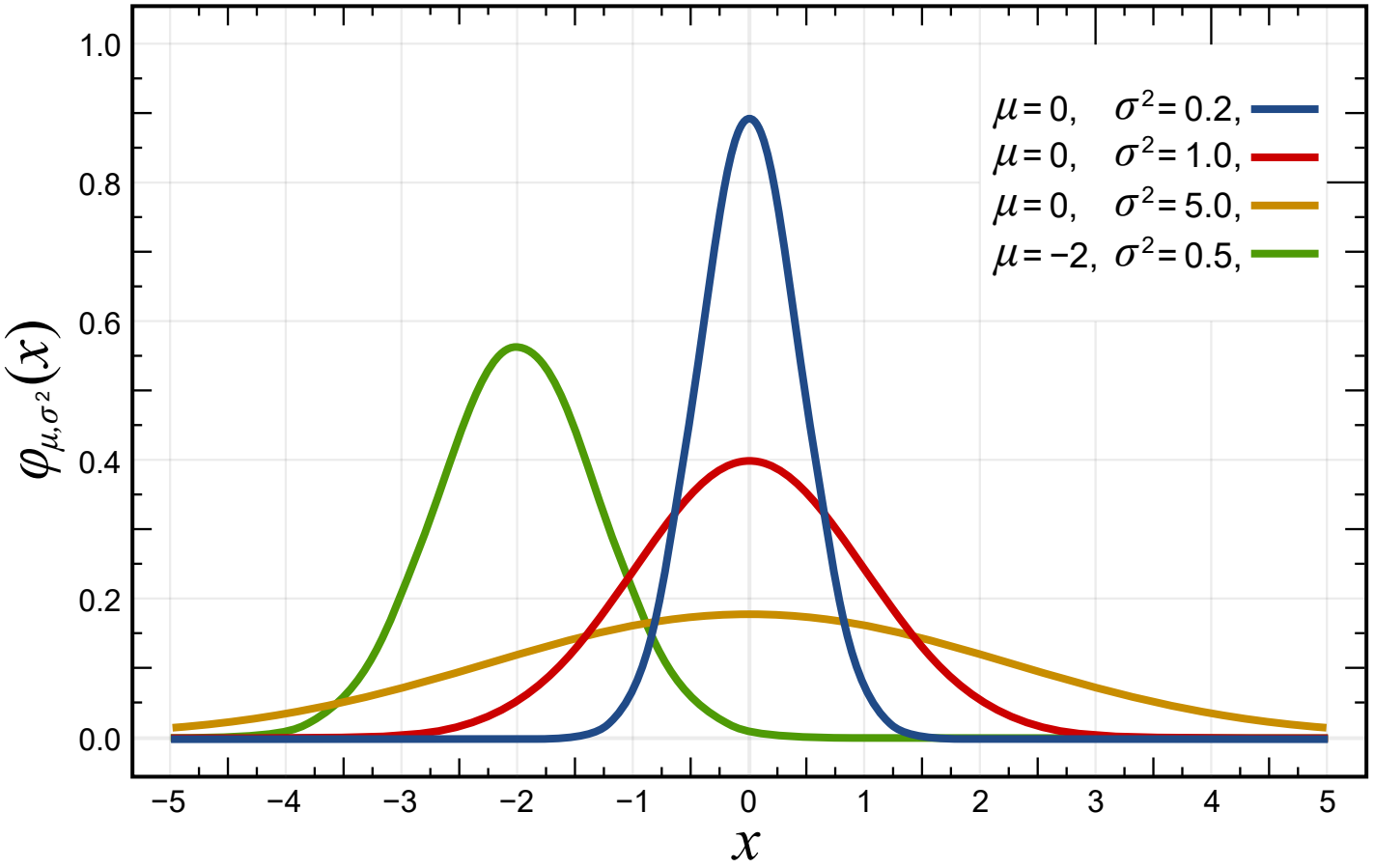
Range = 4

Variance = 2.3

Standard deviation = 1.516

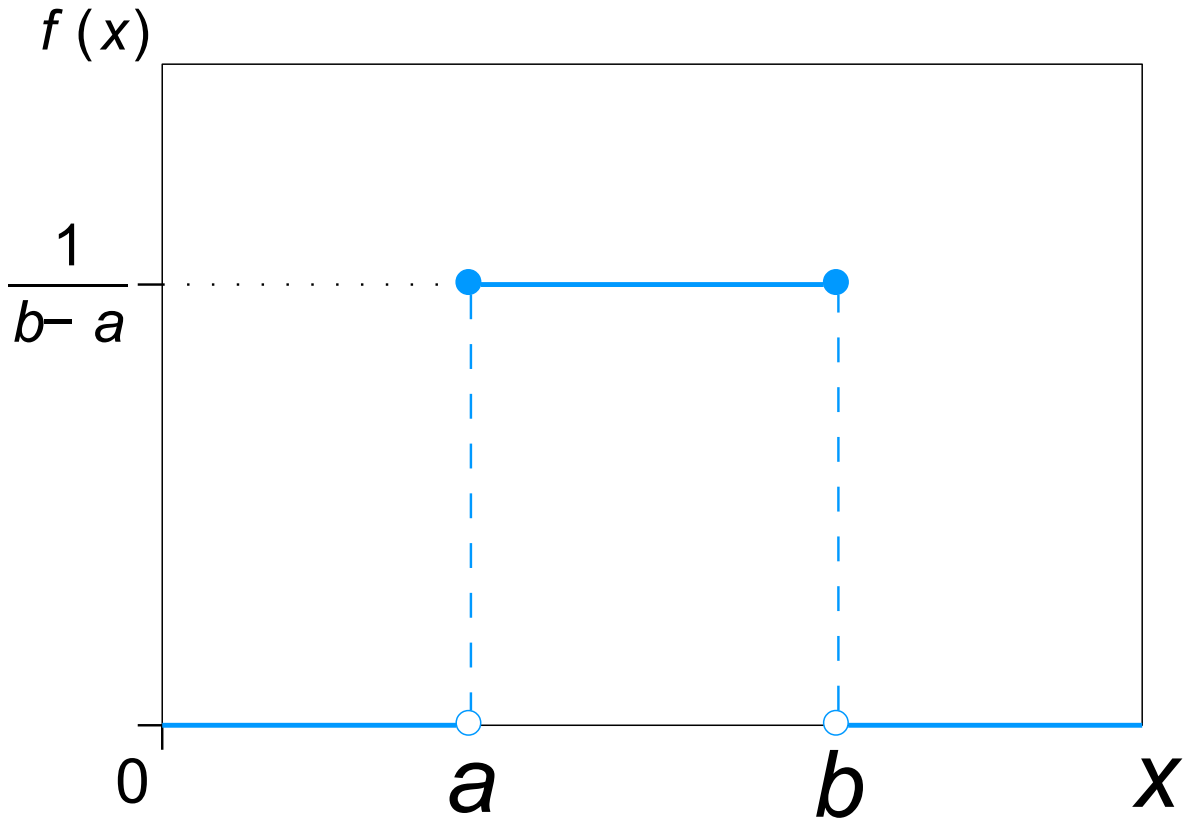
x
7
6
5
6
9

3 Types of Data Distribution - Normal Distribution



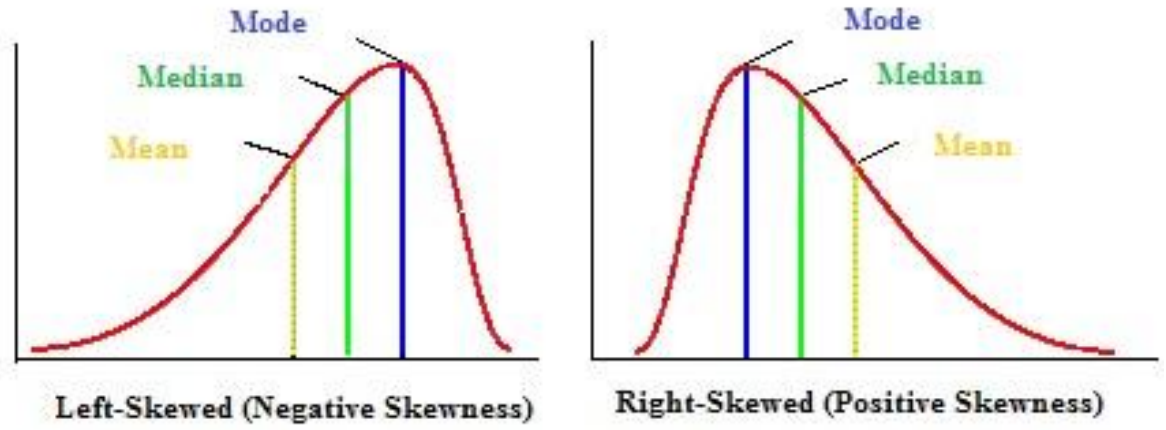
From https://en.wikipedia.org/wiki/Normal_distribution

Uniform Distribution



From https://en.wikipedia.org/wiki/Discrete_uniform_distribution

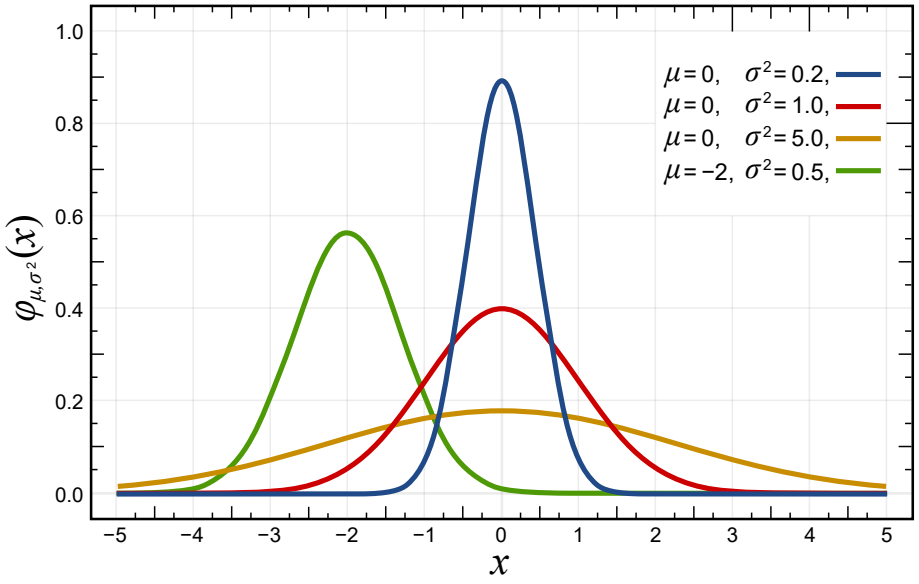
Skewed Distribution



From https://www.statisticshowto.com/probability-and-statistics/skewed-distribution/

Normal Distribution Properties

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ where μ is the population mean and σ is the standard deviation.



Properties of the normal Distribution:

- The mean is the center of this distribution and the highest point.
- The curve is symmetric about the mean. (The area to the left of the mean equals the area to the right of the mean.)
- **The total area under the curve is equal to one.**

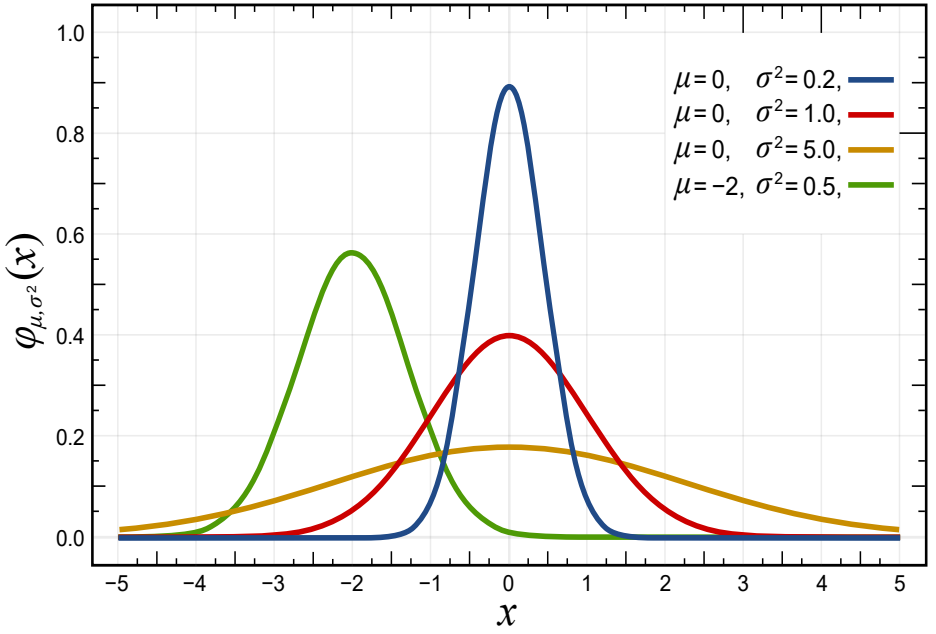
Standard Normal (Z) Distribution

Problem: Unlimited number of possible normal distributions ($-\infty < \mu < \infty$, $\sigma > 0$)

Solution: Standardize the random variable to have mean 0 and standard deviation 1 (z-score).

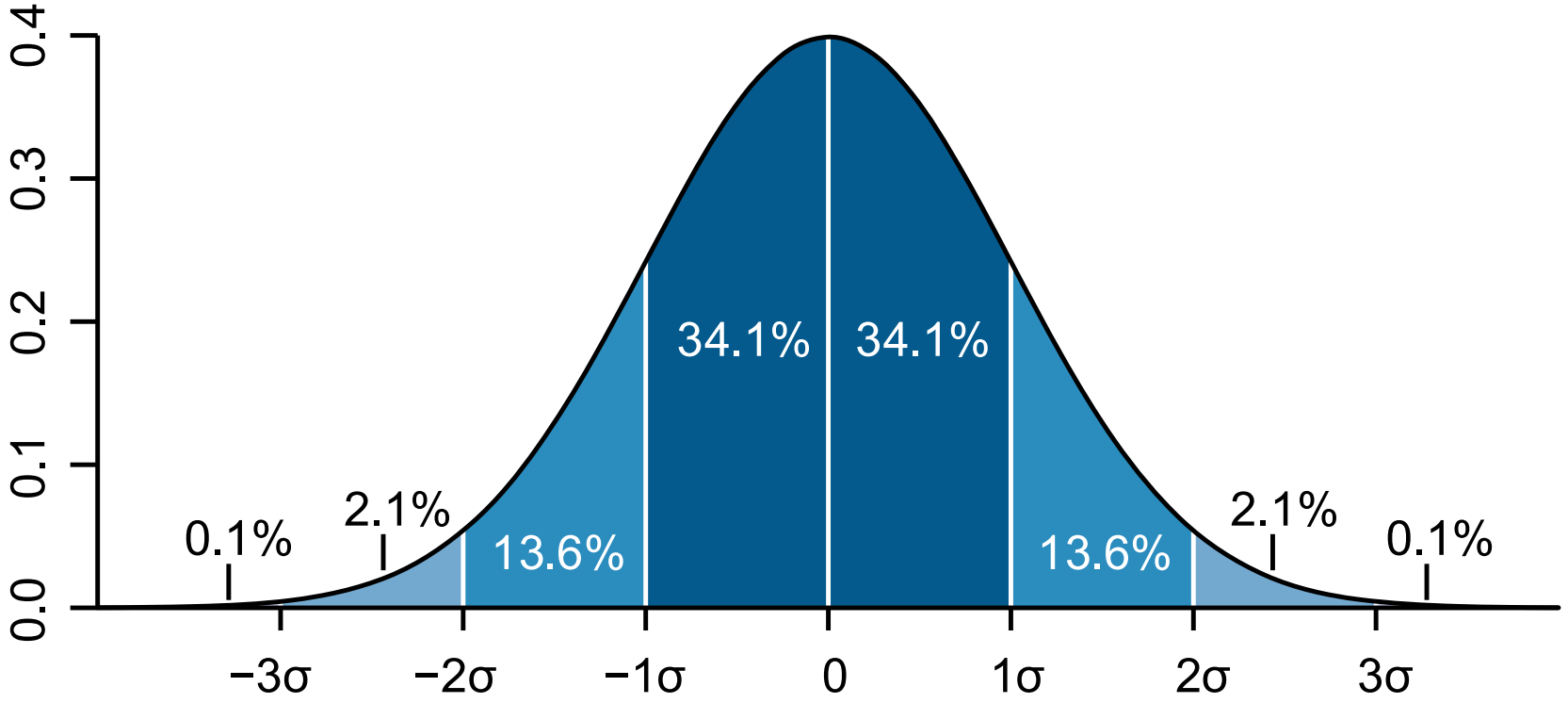
$$Y \sim N(\mu, \sigma) \Rightarrow Z = \frac{Y - \mu}{\sigma} \sim N(0,1)$$

This process is called a standardization, or a z-score.



Standard Normal Distribution

The process of converting a value from a normal distribution to a value for a standard normal distribution is called Standardizing, and requires the use of z-scores.



$P(Y \geq \mu) = 0.50$

$P(\mu - \sigma \leq Y \leq \mu + \sigma) \approx 0.68$

$P(\mu - 2\sigma \leq Y \leq \mu + 2\sigma) \approx 0.95$

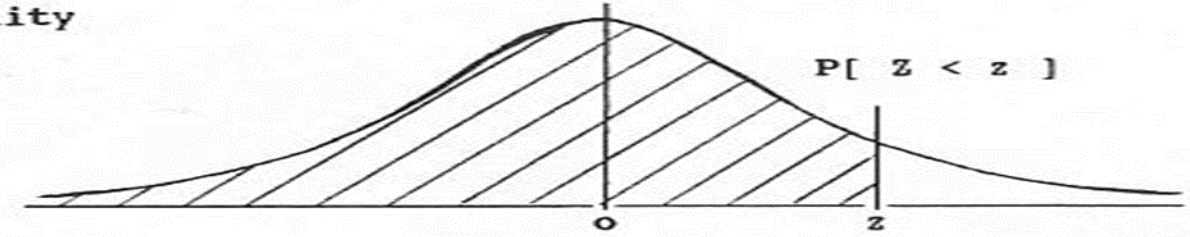
From https://en.wikipedia.org/wiki/Standard_deviation

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[\ Z < z\ ] = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp(-\tfrac{1}{2}Z^2)\ dZ$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
z	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

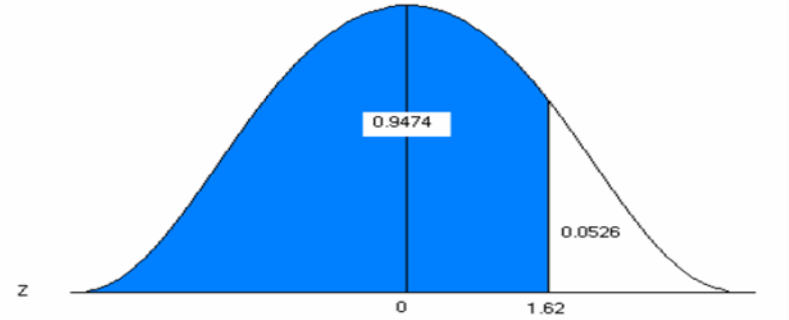
Dr. HalaOwn

Reading the Standard Normal Table

What is the area associated with the Z-score 1.62?

- Read down the Z-column to get the first part of the Z-score (1.6).
- Read across the top row to get the second decimal place in the Z-score (0.02).
- The intersection of this row and column gives the area under the curve to the left of the Z-score.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9595	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633

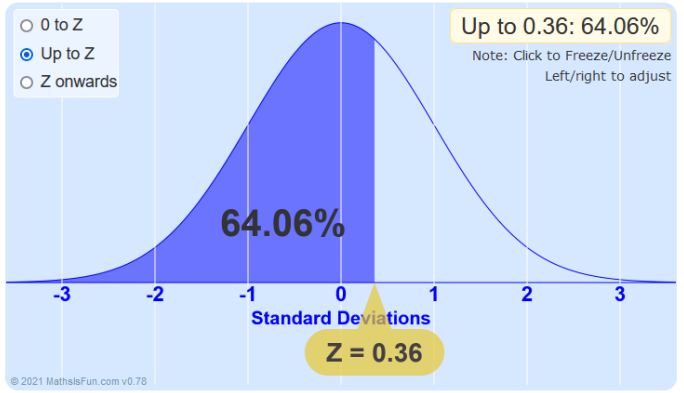


Reading the Standard Normal Table: Example

Find the probability that a variable has a z-score of less than 0.36.

Solution:

Find the z-score in the table. The intersection shows the probability is 64.06%



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133

Normal distribution and probability Example

What's the probability of getting a math SAT score of 575 or less?, Where the mean and s.d. of SAT exam is μ=500 and σ=50?

$$Z = \frac{575-500}{50} = 1.5$$

Remember

$$Y \sim N(\mu, \sigma) \Rightarrow Z = \frac{Y-\mu}{\sigma} \sim N(0,1)$$

i.e., A score of 575 is 1.5 standard deviations above the mean of 500.

Look up Z = 1.50 in standard normal chart, = 0.9332

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

Normal distribution and probability: Example 1

1. Find $P(Z < a)$ for $a = 1.43$
    1. $P(Z \leq 1.43) = F(1.43) = 0.9236$
2. Find $P(Z > a)$ for $a = 1.43$
    1. $P(Z > 1.43) = 1 - F(1.43) = 1 - 0.9236 = 0.0764$

Normal distribution and probability Example 2

Find P(a ≤ Z ≤ b) for a = -1 and b = 1.5

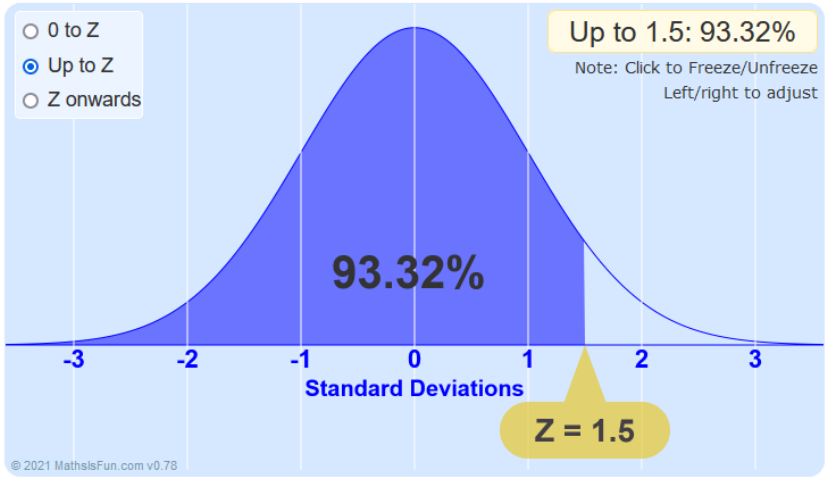
To solve: determine P(b) and P(a), and then subtract.

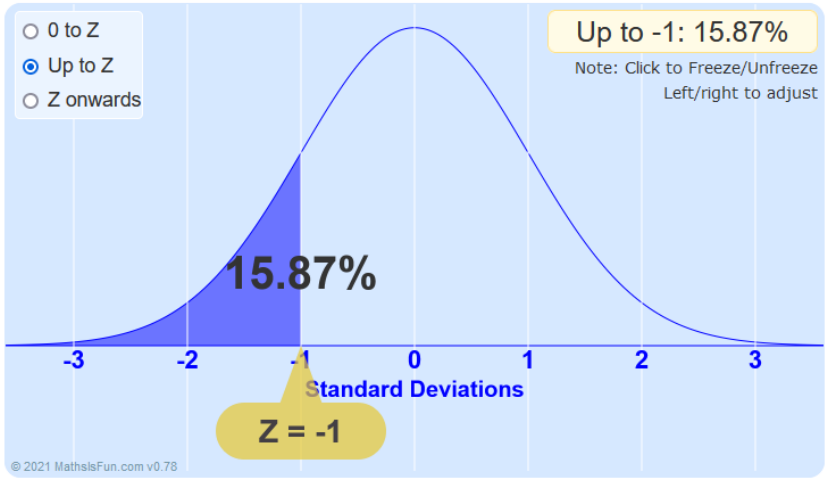
P(-1 ≤ Z ≤ 1.5) =

P(Z < -1) = 0.15866 (the area that we don't want)

P(Z < 1.5) = 0.93319 (the area we do want + what we don't want)

0.93319 - 0.15866 = 0.7745





Application of Z-score: Example

If birth weights in a population are normally distributed with a mean of 109 g and a standard deviation of 13 g, what is the probability of obtaining a birth weight of 141 g or heavier when sampling birth records at random?

What is the probability of obtaining a birth weight of 120 g or lighter?

Application of Z- score Example (cont.)

What is the probability of obtaining a birth weight of 141 g or heavier when sampling birth records at random?

First, standardize the data

$$Z = \frac{141 - 109}{13} = 2.46$$

Second, a z score of 2.46 corresponds to a right tail (greater than) area of: P(Z ≥ 2.46) = 1 - (0.99305)= 0.0069 or .69 %

Application of Z- score Example (cont.)

What is the probability of obtaining a birth weight of 120 or lighter?

First, standardize

$$Z = \frac{120 - 109}{13} = 0.85$$

Second, from the z score table, a value of 0.85 corresponds to a left tail area of:

$P(Z \leq 0.85) = 0.8023 = 80.23\%$

Your turn

Assume a normal distribution IQ test with $\mu = 100$ and $\sigma = 15$, What is the probability of randomly selecting an individual with an IQ score less than 130.

Your Turn Solution

First, convert data to standard normal distribution

$$Z = \frac{130 - 100}{15} = 2.0$$

P(x < 130) = P(Z < 2.0) = 0.9772

Exercise

Adult deer population weights are normally distributed with μ = 110 lb. and σ = 29.7 lb. Biologists determine that a weight less than 82 lb. is unhealthy and you want to know what proportion of your population is unhealthy.

Solution

P(x < 82)= P(Z < -0.94) = 0.1736 $z = \frac{82 - 110}{29.7} = -0.94$

The Z value of 82 is 0.94 standard deviations below the mean.

Approximately 17.36% of the adult deer population is underweight.

Application of Z- score : Standardize Collected Data

Mai score is 680 on the SAT mathematics test. The distribution of the SAT scores are normally distributed with a mean of 500 and a standard deviation of 100.

James take the ACT mathematics test with score 27. The ACT scores are normally distributed with a mean of 18 and a standard deviation of 7.

Who has the higher score?

Why we Standardize our data, cont.

$$Y \sim N(\mu,\sigma) \quad \Rightarrow \quad Z = \frac{Y-\mu}{\sigma} \sim N(0,1)$$

Mai	James
μ=500 and σ=100	μ=18 and σ=6
$Z = \frac{680-500}{100} = 1.8$	$Z = \frac{27-18}{6} = 1.5$

Mai score is higher than James score

Summary

We use descriptive statistics to explore collected data

Normal distribution

Standard normal distribution

Z- score

Applications of Z- score